# Lab 1: Linux C and gcc

In this lab, you will use gcc to write C programs in your Linux environment. There are many different codes to write to discover the Linux terminal, gcc, and the C language. In this lab, you MUST create a directory "lab1/" and put all your files there. You **DO NOT** need to present this lab.

0) Install gcc and the necessary dependencies on your Lubuntu Linux

In a terminal, run the "**sudo apt update**" command. "sudo" means "super user do" because you need the admin rights when you update and install new packages on your Linux distribution.

Run "**sudo apt upgrade**", then "**sudo apt install build-essential**".

If you try "**gcc --version**", you should obtain the version of gcc installed on your system:

```
os2025@os2025-virtualbox:~/c-code$ gcc --version
gcc (Ubuntu 13.3.0-6ubuntu2~24.04) 13.3.0
Copyright (C) 2023 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

os2025@os2025-virtualbox:~/c-code$
```

1) C programs

   a) Hello World!

To start with, just create a C program source "lab1-01-hello-world.c" that will print ---function printf() from "stdio.h"--- the message "Hello, World!" to the terminal when run (with a newline at the end). Compile it with gcc and call the executable "lab1-00-hello", run it and check it works.

   b) Tests:

Create a new C source code in a file "lab1-02-tests.c". In this file, ask the name of the user using scanf(), and the age of the user (both are strings). Convert the age to an integer using the atoi() function ---from "stdlib.h"---.

Then, print a message "Hello, Jerome. You are abc...   :)" where "Jerome" is the name entered by the user and abc is a description depending of the age:

0-12: abc="a child"

13-19: abc="a teenager"

20-30: abc="an adult"

31-60: abc=”experienced”

61+: abc=”wise”

Compile your code with gcc, run it and check the result depending on the entered name and age.

c) Loops

Create a new C source code in a file “lab1-03-loops.c”. In this file, create:

- A for loop that prints all the integers from 0 to 10 (both included), with increment of 1, separated by /: 0/1/2/3//5/6/7/8/9/10 (be careful, we don’t want the / after 10),
- A for loop that prints all the integers from 23 to 7 (both included), with decrement of 1, separated by $: 23$22$21$20$19$18$17$16$15$14$13$12$11$10$9$8$7 (be careful, we don’t want the last $ after 7),
- A for loop that prints all the multiples of 7 between 0 and 123 (123 excluded), separated by space: 0 7 14 21 ... 119
- A while loop that prints the even numbers between 0 and 12 (both included), separated by ; (be careful, we don’t want the ; after 12): 0;2;4;6;8;10;12
- A while loop that prints the odd numbers between 123 and 111 (both included), separated by space: 123 121 119 117 115 113 111

d) Libraries and dynamic library

Create a new C source code in a file “lab1-04-mylib.c” that will contain two functions: The first function printDateToday() will print todays date (an example is given in a lecture). The second function printHiNameAge(char * name, int age) will print the name and age of a person: “Hi Jerome, you are  25.”. If “Jerome” and “25” are passed as parameters to this function. There is no main() function in “lab1-04-mylib.c”, and it is normal, we will not run it but use it in another code. Compile this file as a dynamic library (libmylib.so) and copy it (with sudo) into “/usr/local/lib/”. Add this new library to the default libraries using sudo ldconfig -v”.

Create a new C source code in a file “lab1-05-using-mylib.c” that will execute the two functions from your library “libmylib.so”. compile it and link it to your dynamic library with the “-lmylib” parameter in your gcc command line. You should obtain today’s date and the name and age of a person you called the function with, for instance: printHiNameAge(“Jerome”, 25);

**This is the end. In this lab, you learned to program in C using datatypes, tests, loops, functions, and libraries. Congratulations, grattis!**